TEXORA · SESSIONAL NOTES · STUDY SHEET

Plain Sewing Machines and Practices

Getting to know the plain (single-needle lockstitch) machine before it is studied as a stitch-former: its head parts and their functions, the threading path, needle and thread basics, and the practice exercises that build control of the machine.

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 6 · LEVEL 2 TERM 2

1. Aim and place of the study

Aim: to identify the parts of a plain sewing machine head, state the function of each, thread the machine correctly, and carry out the standard practice exercises — sewing straight lines, curves and corners first on paper without thread, then on fabric. This study is machine familiarity and control; the formation of the lock stitch itself, and the production of stitch samples, is Study 7.

2. The machine head and its parts

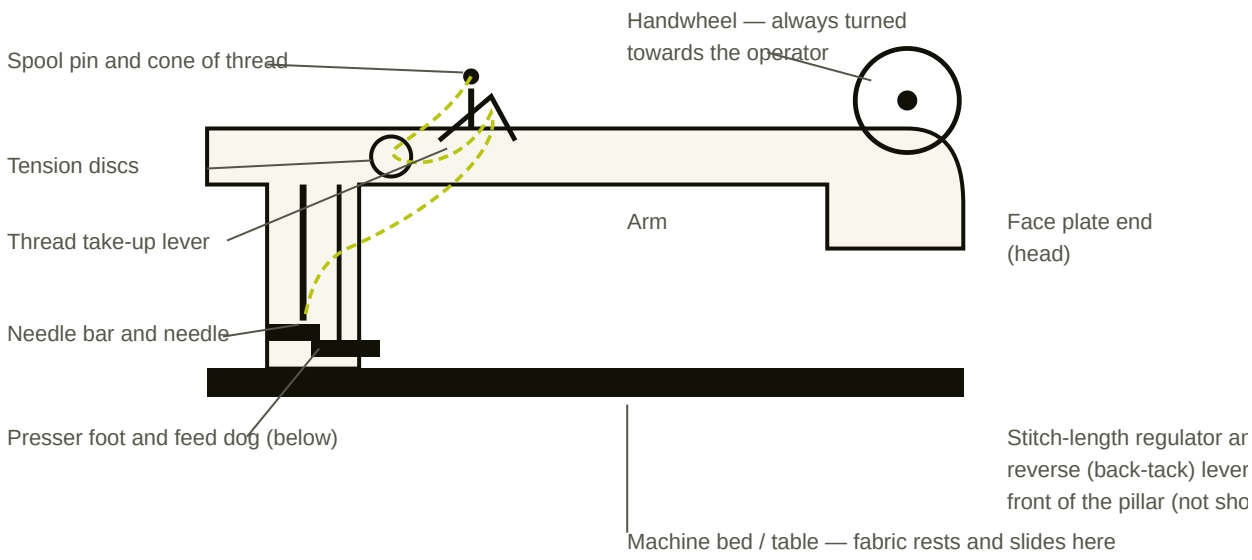


Figure 1 — The plain lockstitch machine head in profile, with the upper threading path shown in green (dashed). The bobbin and hook sit below the needle plate, inside the bed.

PART	FUNCTION
Spool pin / thread stand	Holds the thread cone or spool and delivers thread smoothly to the first guide.

Tension discs and regulator	Two discs press the thread between them; the numbered dial sets the pressure, and so the needle-thread tension (balanced in Study 7).
Thread take-up lever	Rises to pull the stitch tight and falls to supply thread to the needle; the machine is always threaded and started with the lever at its highest point.
Needle bar and needle clamp	Carries the needle up and down; the clamp fixes the needle with its long groove facing the threading side.
Presser bar and presser foot	Holds the fabric against the feed dog; the foot's pressure is adjustable, and the knee lifter or hand lever raises it.
Feed dog	The toothed block under the needle plate that rises, moves the fabric by one stitch length, and drops — the machine's only fabric transport on a plain machine.
Needle (throat) plate and slide plate	The plate carries the needle hole and feed-dog slots and seam-guide markings; the slide plate gives access to the bobbin.
Handwheel	Turns the main shaft by hand for positioning; always rotated towards the operator — turning it backwards tangles thread and can jam the hook.
Stitch-length regulator	Sets the feed movement per stitch — the stitch length, measured as stitches per inch (SPI) in Study 7.
Reverse lever	Reverses the feed for back-tacking at seam starts and ends, locking the stitching.
Bobbin winder	Fills bobbins evenly from the cone while sewing continues; an unevenly wound bobbin gives uneven lower tension.

3. Threading the machine

1. Raise the take-up lever to its highest point by turning the handwheel towards you.
2. Place the cone on the stand; lead the thread through the stand guide to the machine's first thread guide.
3. Pass the thread down and between the tension discs (it must sit between the discs, not over them), around the tension check spring, and up through the eye of the take-up lever.
4. Lead it down through the lower guides to the needle-bar guide, then through the needle eye from the side of the long groove (left to right on most plain machines — confirm on the laboratory machine and record it).
5. Draw about 10 cm of thread under and behind the presser foot.
6. For the lower thread: wind the bobbin evenly, place it in the bobbin case so the thread unwinds in the direction the case requires, draw the thread through the case spring slot, set the case on the

hook's centre pin until it clicks, and pull the bobbin thread up through the needle plate with one handwheel turn while holding the needle thread.

A machine threaded with the take-up lever down, or with the thread over — not between — the tension discs, will sew a few loose stitches and jam. Threading errors are the first thing to check for every fault in this study and the next.

4. Needle and thread basics

- **Needle system:** plain lockstitch machines of the laboratory take the DB×1 system (also marked 16×231). The system letter fixes the shank and length dimensions; a needle of the wrong system cannot be fitted correctly at all.
- **Needle size:** stated in Nm (the blade diameter in hundredths of a millimetre): Nm 70–80 for light fabrics, 90 for medium shirting, 100–110 for heavy material. The rule: the finest needle that carries the thread without breaking it, so the hole left in the fabric is as small as possible.
- **Needle condition:** a bent, blunt or burred needle is replaced, never endured — it causes skipped stitches, thread breaks and fabric damage (Study 7).
- **Thread:** sewing threads are sized by ticket number — the higher the ticket, the finer the thread (a common shirting thread is ticket 100–120 core-spun or spun polyester). Needle thread and bobbin thread are normally the same type and size so the tensions balance.
- **The pairing rule:** thread must pass freely through the needle eye — as a guide, the thread should fill roughly half the eye; too thick a thread in a fine needle shreds at the eye, too fine a thread in a coarse needle gives loose, uncontrolled stitches.

5. The practice exercises

1. **Paper, no thread, needle fitted.** Rule straight parallel lines on paper and sew along them with the needle punching the holes: the exercise teaches straight guiding and even speed without thread faults confusing the lesson. Then ruled squares and triangles (pivoting at corners with the needle down), then circles and spirals (continuous gentle guiding).
2. **Speed control.** Practise starting and stopping exactly on a mark: the treadle or pedal is controlled by the whole foot, and the machine is stopped by heel pressure — never by grabbing the handwheel of a running machine.
3. **Fabric, threaded.** Two plies of exercise fabric: straight seams at a set distance from the edge, using the needle-plate guide lines; back-tack at both ends with the reverse lever; corners pivoted with the needle down.

4. **Judging the result:** lines are checked for straightness against the rule, corners for a single pivot point (not a curve), and stitching for even length — the samples are kept in the sessional record, dated, in exercise order.

6. Precautions and safety

- Fingers never within the needle zone while the machine can run; the needle guard stays fitted.
- Change needles and bobbins with the power off or the motor clutch disengaged.
- Turn the handwheel only towards the operator.
- Loose clothing, ties and hair are kept away from the handwheel and belts; no pins held in the mouth — pins live in the tray.
- Report a bent or broken needle at once; account for every fragment of a broken needle before sewing resumes (a fragment left in work is a wearer injury waiting).

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Name the parts, in order, through which the needle thread passes from cone to needle.
2. Why must the take-up lever be at its highest point when threading and when starting to sew?
3. A student threads the machine but the thread has passed over the tension discs instead of between them. What sewing fault follows, and why?
4. State the needle system of a plain lockstitch machine and the meaning of needle size Nm 90.
5. Why are the first practice exercises sewn on paper without thread?
6. How is a corner sewn sharply, and what single action makes it possible?
7. Why is the handwheel always turned towards the operator?

Answer key

1. Stand guide → machine thread guide → tension discs (between them) and check spring → take-up lever eye → lower guides → needle-bar guide → needle eye.
2. At its highest point the lever has completed pulling up the previous stitch and presents its eye where the thread can be passed; starting from this position prevents an immediate thread tangle and ensures the first stitch is pulled up correctly.
3. No tension is applied to the needle thread, so stitches form as loose loops on the underside (or the thread snarls under the plate) — the discs can only grip and meter thread that passes between them.
4. DB×1 (16×231). Nm 90 means a blade diameter of 90/100 mm = 0.90 mm.

5. To isolate the skill being taught — guiding and speed control — from thread behaviour; the punched holes give a permanent, measurable record of accuracy, and no fabric or thread is wasted while control is poor.
6. The machine is stopped with the needle down in the fabric at the corner point, the presser foot is raised, the work is pivoted around the needle to the new direction, the foot is lowered and sewing continues — the needle-down pivot makes the sharp corner.
7. The machine's timing — hook rotation, feed motion and take-up action — is designed for that rotation; turning backwards drives the hook against the thread and needle, causing tangles, jams and possible damage.

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